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Comparative Analysis of Self-Supervised Learning Frameworks for Vision-Based PM2.5 Air Quality Estimation

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Comparative Analysis of Self-Supervised Learning Frameworks for Vision-Based PM2.5 Air Quality Estimation

1. Abstract

Calculating concentrations of PM2.5 using standard photos provides an economical and scalable alternative to building a costly network of physical sensors. However, the traditional method of training supervised deep learning models involves collecting vast amounts of labeled data, which poses challenges with respect to air quality indexes (AQI). In this study, the efficacy of three self-supervised learning architectures – BYOL, Barlow Twins, and DINO – is investigated against a strong supervised model. All of the models were tested using the PM25Vision dataset, consisting of 11,219 photographs classified according to their severity level, which ranged from 0-50 to 301-600. Although ResNet-50 attained the best performance with a supervised accuracy of 61.41% using a 90:10 split, EfficientNet-B0, which recorded a supervision accuracy of 60.34%, was chosen for both self-supervised architectures because it is efficient to train on Kaggle GPUs.

In the SSL frameworks, DINO performed much better than the non-distillation frameworks, scoring 45.50% and 68.36% for linear probe accuracy and fine-tuned accuracy, respectively. Most importantly, the proposed framework exhibited an outstanding ability to learn from very few training labels, as it was able to score over 30% accuracy with just 1% labeled training data. More importantly, it was successful in detecting the most extreme and dangerous haze types (Class 301-600), for which BYOL and Barlow Twins failed entirely. Statistical significance testing validates that DINO is better (Friedman p=0.0011).

2. Introduction

Air pollution is a serious and growing problem around the world. Particulate matter of size 2.5 micrometers (PM2.5) can cause several respiratory ailments, cardiovascular disorders, and rising death rates across the globe. Conventional monitoring of AQI utilizes hardware devices stationed physically on the ground. Though they provide high accuracy, their deployment costs are rather high. It leads to limited spatial coverage; therefore, such devices do not operate efficiently in rural and underdeveloped locations due to unavailability and non-affordability.

In order to overcome the above monitoring problem, visual-based estimation of PM2.5 becomes an affordable solution. This method exploits the physics of scattering, contrast loss, and color degradation present in conventional digital photos for estimating ambient air quality.

Figure 1: Visual contrast between severe PM2.5 haze obscuring background landmarks (right) and clear, low-AQI conditions with high visibility (left).

Despite its significant potential, however, the biggest limitation in using vision-based approaches to estimate AQI is the extensive use of very large and precisely labeled datasets. Acquiring ground truth data for PM2.5 concentration for each image requires referencing to the physical sensor stations active in the same place and at the same time as the photo was taken. As a result, the performance of supervised learning approaches tends to suffer from overfitting, which limits their ability to adapt to new locations.

On the other hand, Self-Supervised Learning (SSL) provides an interesting avenue for tackling the annotation bottleneck. By extracting generalizable features from the intrinsic structure of unlabeled visual data, SSL significantly lowers the dependency on extensive and costly labeling efforts. This work attempts to explore the potential of modern SSL frameworks in detecting useful signals to classify images according to the level of severity regarding PM2.5 concentration in the atmosphere. Three major SSL paradigms are considered: bootstrapping (BYOL), redundancy reduction (Barlow Twins), and self-distillation (DINO).

3. Related Work

There is a move from physical air quality monitoring to vision-based estimations which have been enabled by deep learning algorithms used to infer the degradation in atmospheric scattering, contrast reduction, and color shifting due to PM2.5 concentration. Whereas earlier computational algorithms employed mathematically intelligible yet brittle optical properties such as dark channel priors, image entropy, and transmission map estimation, more recent research focuses on using end-to-end Convolutional Neural Networks (CNNs) for estimating AQIs directly from pixel-level data. Early foundational studies have successfully leveraged CNNs with crowd-sourced landscape photos to show that deep architectures can learn complex, non-linear functions of degradation and pollution levels from visual data across varying urban environments. To leverage time dynamics in addition to spatial feature extraction, hybrid architectures have grown more popular in recent times. For example, hybrid architectures that use ResNet as a feature extractor in combination with LSTMs to analyze time sequences of

smartphone image data have become common practice. In the same way, estimation systems built on mobile devices have merged with lightweight efficient Net backbones and Bidirectional LSTMs, and they show that harnessing the power of both HDR images taken by smartphones and non-HDR images, along with additional information regarding weather, makes precise PM2.5 estimations possible while keeping the computational cost low. There are other methods designed to refine the process of regression; the proposed ensemble models of CNN and Random Forest make use of the CNN solely for the deep learning-based extraction of features like edge degradation and color fading and leave the task of predicting AQIs in the hands of the Random Forest algorithm, which resulted in up to 11.11% decreased MAE owing to overfitting prevention. Despite all of these innovations in architectures, supervised machine learning models suffer from the lack of geographically dispersed data and overfit in terms of specific landmarks. To counter this dire shortage of training data, recent studies have investigated Few-Shot Learning approaches based on Prototypical Networks that learn a metric space wherein images are classified by measuring the distance to prototypes representing each AQI category, retaining more than 92% accuracy while requiring very few labeled examples. Further pushing the boundaries of label efficiency, the most promising approach today is moving towards Self-Supervised Learning (SSL), enabling the extraction of universal atmospheric representations from copious unlabelled images. Early contrastive learning approaches such as SimCLR and MoCo made groundbreaking advances in unsupervised feature learning by forcing augmented images to converge. However, their dependence on massive batch sizes makes them impossible to run on edge devices. Thus, attention has shifted to non-contrastive approaches, such as BYOL, which uses the exponential moving average (EMA) momentum encoder to avoid representation collapse without the need for negative samples, and Barlow Twins, which works by reducing the redundancy of cross-correlation between the two symmetrically augmented views mathematically. Moreover, self-distillation methods, including DINO, can be seen as a pivotal advance in atmospheric studies; by using the multi-crop technique, which makes the student model learn the overall atmospheric context based on small pixel regions, DINO is capable of handling the dispersed and unstructured characteristic of dense smog, providing an effective approach for PM2.5 monitoring.

4. Methods

4.1. Exploratory Data Analysis and Preprocessing

Before training the model, an extensive exploratory analysis of the PM25Vision dataset was performed in order to evaluate the quality of the data and develop the augmentation pipeline. Histograms for the RGB and HSV color spaces were computed to examine brightness, contrast, and saturation, key factors that reveal the effects of atmospheric scattering. The sharpness and structural noise of images were rigorously estimated by evaluating the Laplacian variance, while a gray world test was carried out to assess the sanity of the camera white balance.

For stringent protection against any data leaks between the train and test sets, all perceptually similar and duplicate images were identified and separated using perceptual hashing techniques,

along with proper group protections during balancing. An empirical study was then performed on image transformations to distinguish between benign transformations that preserve the required labels and harmful transformations that obliterate the essential color gradient patterns and scattering patterns. The study found that while spatial cropping and flipping were relatively harmless, excessive color jittering and excessive blurring completely obliterated the essential PM2.5 haze properties.

4.2. Supervised Baselines

For setting up the SSL baseline and choosing the best performing backbone network, four benchmark supervised convolutional networks were used: ResNet-50, GoogLeNet (Inception-V3), MobileNetV2, and EfficientNet-B0. In order to evaluate robustness with respect to different amounts of training data, all networks were trained on the 9-ratio split grid from 90:10 (train:test) split ratio to 10:90.

Table 1: ResNet-50 Performance

Split	Images (Train/Val/Test)	Accuracy	Macro F1
10:90	1,009 / 112 / 10,098	**45.41%**	0.43
20:80	2,019 / 224 / 8,976	**48.64%**	0.46
30:70	3,029 / 336 / 7,854	**48.40%**	0.46
40:60	4,038 / 449 / 6,732	**51.81%**	0.50
50:50	5,048 / 561 / 5,610	**54.99%**	0.53
60:40	6,057 / 674 / 4,488	**57.78%**	0.56
70:30	7,067 / 786 / 3,366	**57.07%**	0.55
80:20	8,077 / 898 / 2,244	**58.11%**	0.56
90:10	9,087 / 1,010 / 1,122	**61.41%**	0.60

Table 2: EfficientNet-B0 Performance

Split	Images (Train/Val/Test)	Accuracy	Macro F1
10:90	1,009 / 112 / 10,098	**46.12%**	0.45
20:80	2,019 / 224 / 8,976	**49.69%**	0.48
30:70	3,029 / 336 / 7,854	**51.52%**	0.50
40:60	4,038 / 449 / 6,732	**52.38%**	0.51
50:50	5,048 / 561 / 5,610	**54.55%**	0.53
60:40	6,057 / 674 / 4,488	**54.90%**	0.54
70:30	7,067 / 786 / 3,366	**57.04%**	0.56
80:20	8,077 / 898 / 2,244	**59.31%**	0.58
90:10	9,087 / 1,010 / 1,122	**60.34%**	0.59

Table 3: GoogLeNet (Inception-V3) Performance

Split	Images (Train/Val/Test)	Accuracy	Macro F1
10:90	1,008 / 113 / 10,098	**45.41%**	0.43
20:80	2,018 / 225 / 8,976	**43.76%**	0.41
30:70	3,028 / 337 / 7,854	**48.17%**	0.45
40:60	4,038 / 449 / 6,732	**50.62%**	0.48
50:50	5,048 / 561 / 5,610	**51.27%**	0.49
60:40	6,057 / 674 / 4,488	**55.08%**	0.53
70:30	7,067 / 786 / 3,366	**56.54%**	0.54
80:20	8,077 / 898 / 2,244	**58.73%**	0.57
90:10	9,087 / 1,010 / 1,122	**60.34%**	0.59

Table 4: MobileNetV2 Performance

Split	Images (Train/Val/Test)	Accuracy	Macro F1
10:90	1,008 / 113 / 10,098	**45.41%**	0.43
20:80	2,018 / 225 / 8,976	**48.33%**	0.46
30:70	3,028 / 337 / 7,854	**48.40%**	0.46
40:60	4,038 / 449 / 6,732	**51.81%**	0.50
50:50	5,048 / 561 / 5,610	**54.99%**	0.53
60:40	6,057 / 674 / 4,488	**57.78%**	0.56
70:30	7,067 / 786 / 3,366	**58.73%**	0.57
80:20	8,077 / 898 / 2,244	**59.31%**	0.58
90:10	9,087 / 1,010 / 1,122	**60.34%**	0.59

ResNet-50 exhibited superior performance metrics in the supervised setting, with 61.41% accuracy and 0.60 Macro F1 score on the 90:10 split. EfficientNet-B0 showed very strong performance, achieving 60.34% accuracy and 0.59 Macro F1 score on the 90:10 split. Both GoogLeNet and MobileNetV2 reached their maximum accuracy of 60.34% on the 90:10 split, although EfficientNet-B0 was chosen as the backbone feature extractor for Self-Supervised Learning experiments. The choice was driven by the fact that EfficientNet-B0 provides the perfect combination of representational power and computational efficiency, with just 0.394 GFLOPS needed. EfficientNet-B0 is absolutely essential due to its lightweight nature and the necessity to meet the high GPU memory demands of multi-crop batch training in the Self-Supervised Learning stage.

4.3. Self-Supervised Frameworks

To address the issues associated with the supervised baselines, three different SSL architectures were designed using the common EfficientNet-B0 backbone. Each model was trained without labels for 30 epochs before being used for downstream analysis.

BYOL (Bootstrap Your Own Latent): To enable asymmetrical augmentation techniques that mimic hazy environments, BYOL was used. It consists of an online and a target model. The weights of the target model are generated by an exponentially weighted average (EMA) of the online model's weights, thereby serving as the momentum encoder to guarantee stable representations while avoiding model collapse.

Barlow Twins: Mathematically minimizing the correlation between the two symmetric views, Barlow Twins was designed based on the augmentation probe of EDA. To prevent any disruption to the atmospheric color changes, which are critical for predicting PM2.5 concentration, a minimal ColorJitter augmentation (factor = 0.2) was chosen.

DINO (Self-Distillation with No Labels): Uses a self-distillation approach that adopts a multi-crop augmentation method. The network receives 2 global images (224x224 pixels) and 4 local images (96x96 pixels). The student model is trained to predict the same probability distribution as the teacher model that undergoes momentum update. This makes the student learn to infer the overall atmospheric environment from the local pixels, resulting in a natural spatial hierarchy for haze detection.

5. Experimental Setup

5.1. Dataset and Splitting Strategy

All experimental work was performed on the PM25Vision dataset that consists of 11,219 samples exhaustively annotated based on the six AQI severity levels including 0-50 (Good), 51-100 (Moderate), 101-150 (Unhealthy for Sensitive Groups), 151-200 (Unhealthy), 201-300 (Very Unhealthy), and 301-600 (Hazardous). In order to create a consistent baseline for training the supervised models, we began by splitting the dataset into an 80:20 training-testing set using stratified sampling technique. Additionally, in order to evaluate the stability of our extractors under different scarcity ratios of training data, downstream evaluations were carried out using a 9-ratio range from 90:10 to 10:90.

5.2. Pretraining and Hyperparameters

The BYOL, Barlow Twins, and DINO models were implemented using the common EfficientNet-B0 backbone architecture. For proper comparative evaluation, pre-training without labels for each of these model architectures was done consistently for 30 epochs.

Given the hardware limitations in the Kaggle environment, batch size selection had to take into account the memory usage per sample processed by the model. The BYOL and Barlow Twins models could run efficiently with batch size equaling 32; however, since the DINO model required six different views of each image, the batch size for this model was lowered to 16 to avoid running out of memory.

Table 2: SSL Pretraining Hyperparameter Configuration

Hyperparameter	BYOL	Barlow Twins	DINO
Backbone Architecture	EfficientNet-B0	EfficientNet-B0	EfficientNet-B0
Pretraining Epochs	30	30	30
Optimizer	AdamW	AdamW	AdamW
Learning Rate Schedule	CosineAnnealing	CosineAnnealing	CosineAnnealing
Batch Size	32	32	16
Hyperparameter	**BYOL**	**Barlow Twins**	**DINO**

5.3. Evaluation Protocol

Once the model weights had been frozen post-label-free pre-training, the subsequent evaluation process involved several stages:

Linear probing: As the primary metric for SSL, one linear layer was used to train the classification on top of the feature representations provided by the respective models. In order to validate the quality of the obtained features, several classifiers such as MLP, SVM, Decision Tree, and Random Forest were employed for evaluating the representations.

Full fine-tuning: All the layers of the network were fine-tuned end-to-end in an attempt to achieve better performance for 20 training epochs. The performance gap between linear probing and full fine-tuning was measured.

Embedding visualization: To evaluate how the representation of AQI classes differ in the latent space, dimensional reductions via t-SNE, UMAP, and PCA were used. Cluster density was quantified using the silhouette score, and k-Nearest Neighbor (k-NN) classification accuracy (k €{1, 5, 20}) was measured.

Label efficiency: To simulate the case of sparse labeled data, the linear probing accuracy was calculated for increasing amounts of training data, including 1%, 5%, 10%, 25%, and 50% of all the labeled samples.

6. Results

6.1. Linear Probing and Fine-Tuning Performance

The primary yardstick for evaluating the learned representations was the linear probe, where a single classification layer was trained on top of the frozen SSL encoders.

Table 2: Downstream Performance of SSL Models

SSL Architecture	Linear Probe Accuracy	Full Fine-Tuned Accuracy
BYOL	30.12%	45.05%
Barlow Twins	29.86%	43.32%
DINO	**45.50%**	**68.36%**

As seen in the frozen-feature linear probe evaluation, DINO achieved significantly higher results than any other non-distillation architecture, achieving 45.50% accuracy and macro F1 equal to 0.4753. Meanwhile, both BYOL and Barlow Twins showed a representation collapse on the difficult haze boundaries, obtaining 30.12% and 29.86% accuracy correspondingly.

After full network fine-tuning for 20 epochs, the representation difference between DINO and other architectures was still large. DINO continued to be ahead, obtaining 68.36% accuracy. Meanwhile, the results for BYOL were 45.05%, while for Barlow Twins, they were even lower – 43.32%. It is worth noting that DINO's fine-tuned results (68.36%) significantly exceeded those of the best supervised baseline model (ResNet-50 with 61.41%).

6.2. Per-Class Analysis on Hazardous Conditions

An evaluation of accuracy per class showed a crucial weakness of contrastive and redundancy-reduction approaches in highly degraded visual scenes. Class 301-600 is an example of dangerous haze, wherein the structures have been highly obliterated and only color shadows remain.

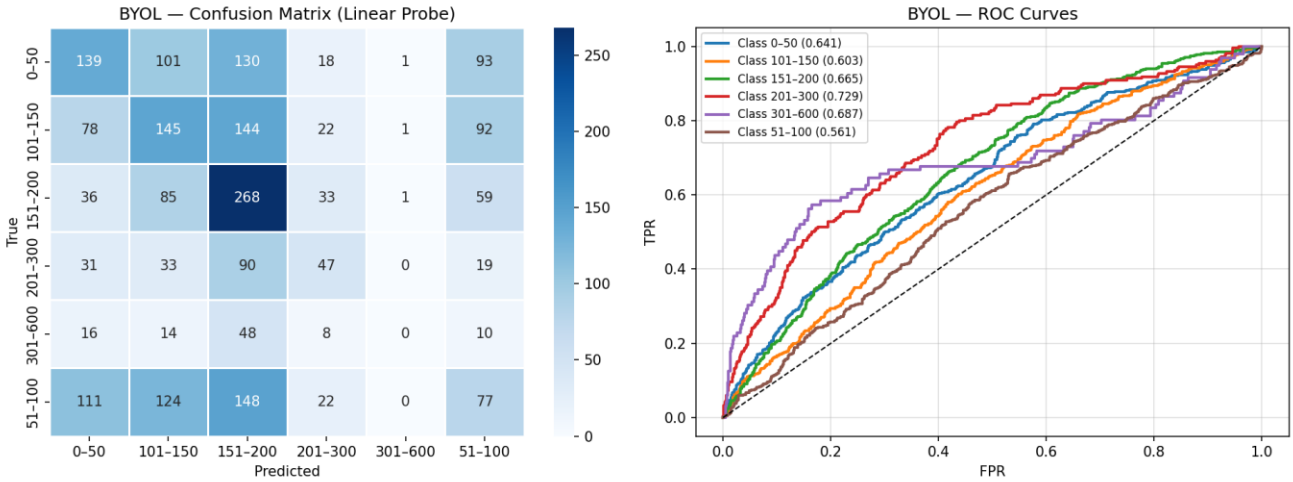

Figure 1: BYOL Confusion Matrix and ROC curve, showing a complete failure (0.00%) to classify the hazardous 301-600 AQI class.

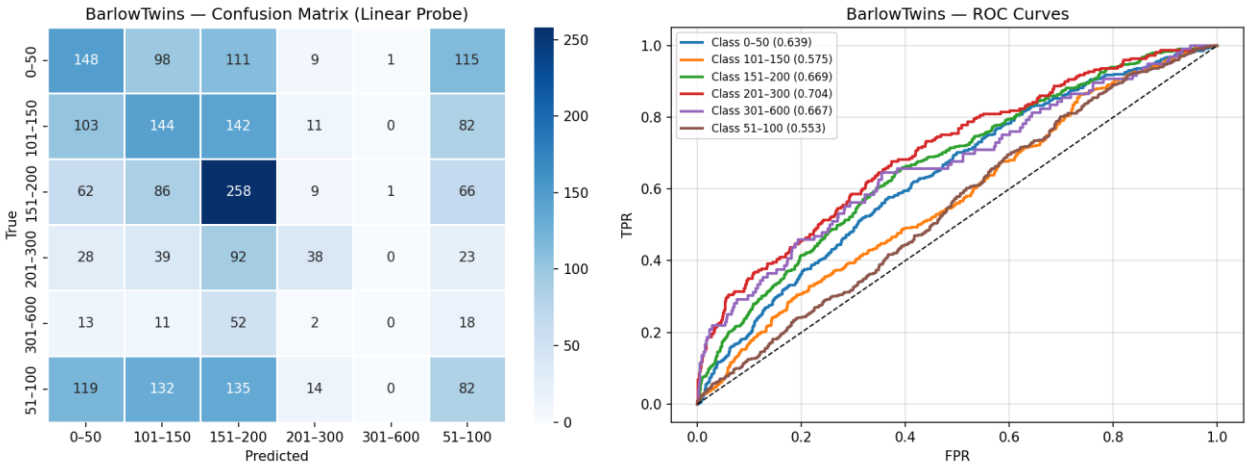

Figure 2: Barlow Twins Confusion Matrix and ROC curve, demonstrating a similar 0.00% detection rate for the extreme 301-600 AQI bin.

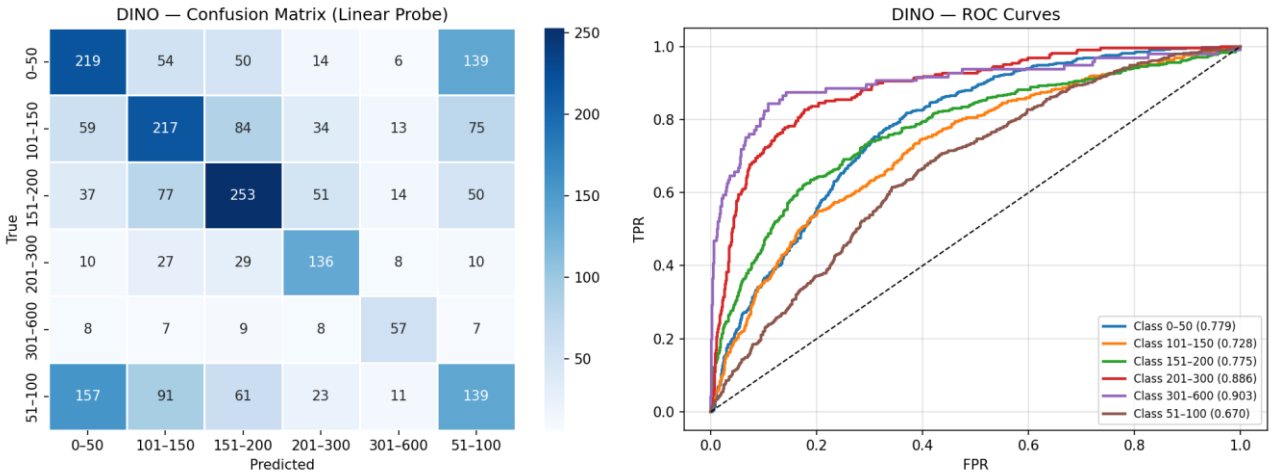

Figure 3: DINO Confusion Matrix and ROC curve, highlighting its success with a 59.38% classification accuracy on the severe 301-600 AQI class.

Under such extreme conditions, both BYOL and Barlow Twins have failed miserably with 0% accuracy on the 301-600 pollution severity bin. On the other hand, DINO has achieved classification of this extreme bin with an accuracy of 59.38%. From the results, it can be inferred that DINO's approach to learning by using multiple crops is effective in learning local textures.

6.3. Label Efficiency

A label efficiency test was performed using a set of active AQI labels that were 1%, 5%, 10%, 25%, and 50% of the total amount of labeled training data for real-life situations.

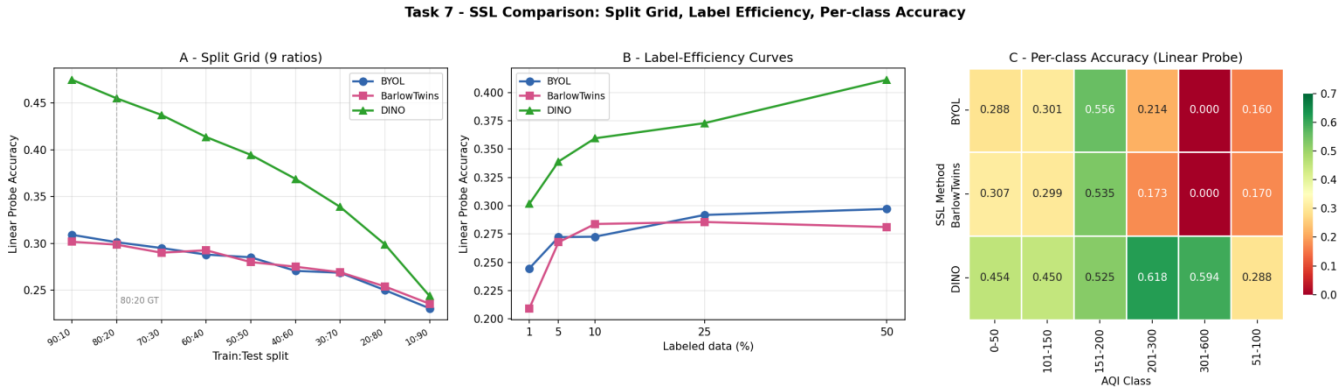

Figure 3: Label-efficiency curve

There were outstanding levels of sample efficiency exhibited by DINO. DINO obtained an accuracy of 30.17% with the utilization of only 1% of the downstream labeled dataset. Notably, the results obtained with only 1% sample of DINO are better than the linear probe accuracy for both BYOL and Barlow Twins with 100% of the data.

6.4. Embedding Space Analysis

To quantify how well the frozen representations separated the AQI classes prior to any labeled training, t-SNE, UMAP, and PCA dimensionality reduction plots were generated.

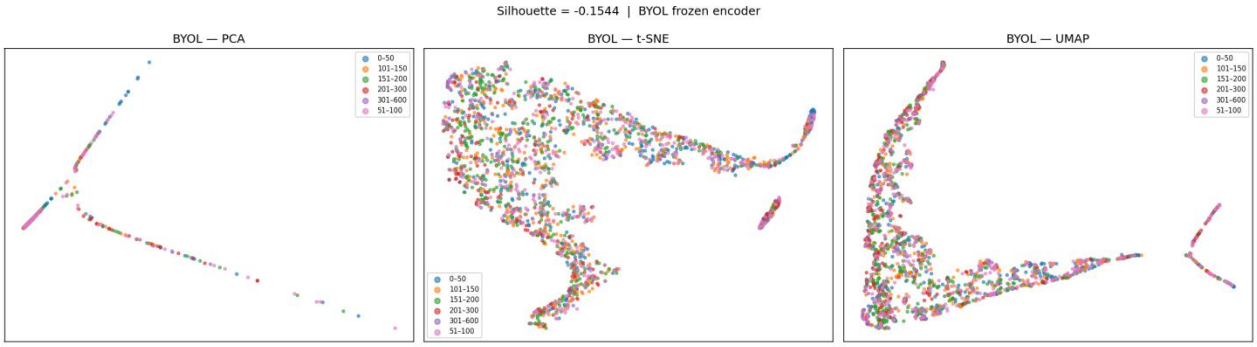

Figure 4: Dimensionality reduction visualizations (PCA, t-SNE, and UMAP) for BYOL

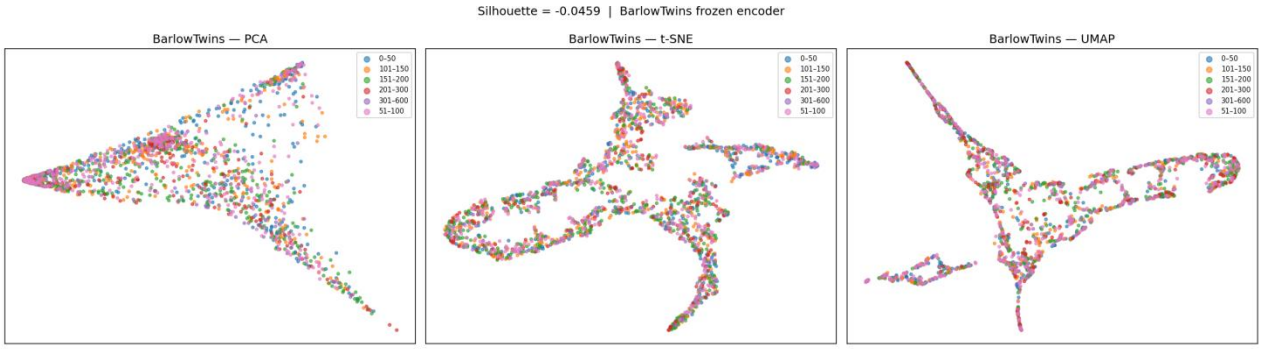

Figure 5: Dimensionality reduction visualizations (PCA, t-SNE, and UMAP) for Barlow Twins

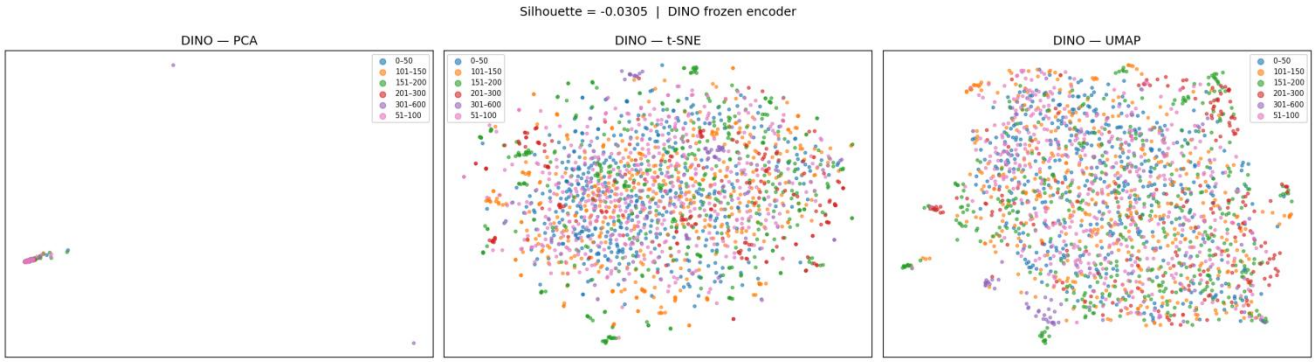

Figure 6: Dimensionality reduction visualizations (PCA, t-SNE, and UMAP) for DINO

Since the haze exhibits continuity visually where the difference between AQI 150 and AQI 151 is negligible, all models resulted in negative Silhouette scores since there was an overlapping cluster. However, haze cannot be referred to as discrete objects such as "cat" and "dog." Although there was continuous overlap, DINO demonstrated the best structural separation with a Silhouette score of -0.0305 in contrast to BYOL (-0.1544) and Barlow Twins (-0.0459).

7. Ablation Studies and Statistical Analysis

To study the sensitivity of the Self-Supervised Learning models towards their key hyperparameters and verify the accuracy of the performance disparities observed after downstream evaluation, an extensive set of ablation studies was performed.

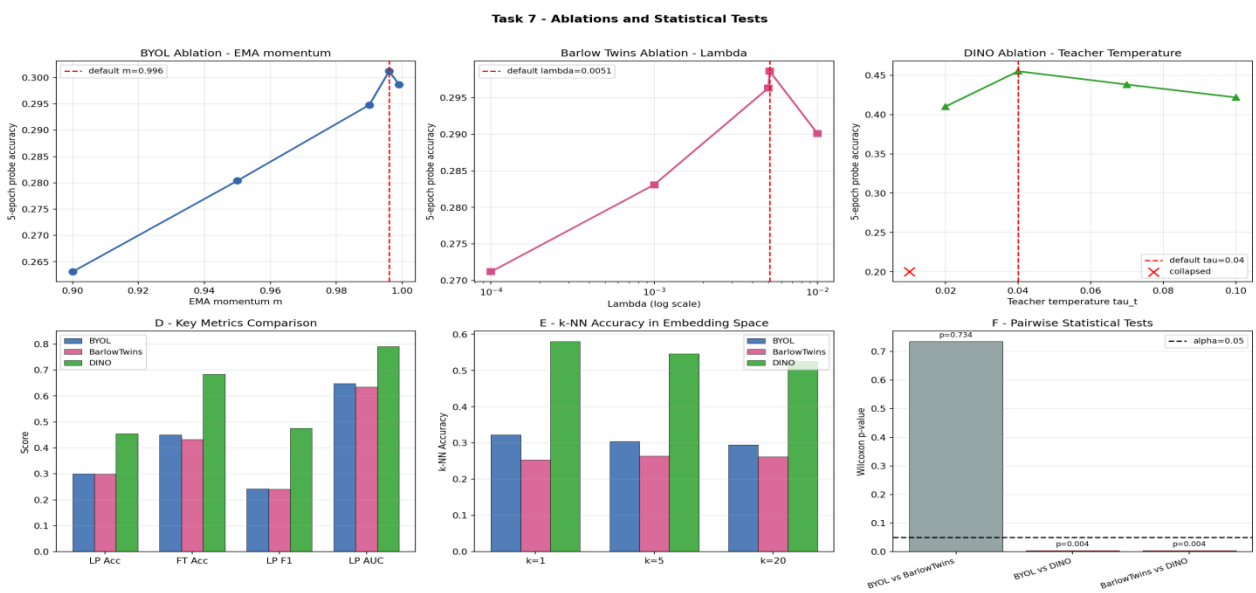

Figure 7: Comprehensive analysis showing SSL hyperparameter ablations, k-NN embedding accuracy comparisons, and statistical validation

7.1. Hyperparameter Ablations

The stability of the SSL method is determined by certain architectural parameters. In order to evaluate the representation quality of the network for different values of the hyperparameters, a 5-epoch linear probe is employed:

BYYOL (EMA Momentum, m): EMA momentum determines how smooth the target model updates its weights. As can be seen from the graph, m=0.996, the default value, provides the best representation. A significant decrease in momentum results in severe degradation of performance and shows that BYYOL is extremely dependent on stability of the target network in order to avoid representation collapse.

Barlow Twins (Redundancy Weight, λ): determines the λ relative importance of the diagonal and off-diagonal components of the cross-correlation matrix. The performance peaks exactly at the default value of λ=0.0051. Any deviations, especially an increase in the value logarithmically, lead to rapid loss in probe accuracy.

DINO (Teacher Temperature,τ): Temperature controls the sharpness of the teacher's output distribution. The default τ =0.04 proved optimal. Crucially, dropping the temperature too low (τ =0.02) resulted in a complete mode collapse, while higher temperatures (τ > 0.04) resulted in overly uniform, less discriminative features.

7.2. k-NN Accuracy in Embedding Space

To test the quality of features without adding any extra parameters like the linear probe, k-nearest neighbors (k-NN) classification was done directly in the embedding space, where k Є {1, 5, 20}. DINO showed a clear supremacy over all k-values, reaching around 60% accuracy when k=1. On the other hand, BYOL and Barlow Twins were in the range of 25%-32% accuracy regardless of k-values. This clearly shows the supremacy of DINO's self-distillation approach to cluster images according to AQI severity.

7.3. Statistical Significance

To determine if DINO's outperformance was statistically significant rather than a byproduct of random weight initialization, pairwise Wilcoxon signed-rank tests were conducted across the experimental splits with an alpha threshold of α=0.05.

BYOL vs. Barlow Twins: The test yielded p=0.734, indicating no statistically significant difference between the two models. Both struggled equally with the continuous nature of the haze dataset.

DINO vs. BYOL & Barlow Twins: The tests for DINO against both BYOL and Barlow Twins yielded a highly significant p=0.004. Because p < 0.05, the null hypothesis is rejected, conclusively proving that DINO's multi-crop self-distillation architecture provides a mathematically significant advantage for vision-based PM2.5 estimation.

8. Discussion and Conclusion

8.1. Discussion

The primary goal of this experiment was to ascertain whether SSL was able to provide a means of representing atmospheric haze in order to estimate PM2.5 values without the need for large, exhaustively labeled data. This paper has proven that the selection of SSL model is of paramount importance when dealing with non-discrete, continuous data such as air pollution.

The undeniable superiority of DINO (linear probe = 45.50%, fine-tuned = 68.36%) over BYOL and Barlow Twins clearly illustrates the benefit of using multi-crop self-distillation. Non-contrastive models such as BYOL and Barlow Twins function based on matching augmented global views. However, in the case of the 301-600 (Hazardous) AQI class, extremely high PM2.5 levels create a visual effect wherein the haze washes out global structures and leaves only uniform color patches. As a result, BYOL and Barlow Twins failed to represent such extreme haze due to their reliance on matching large-scale structures, leading to a complete representation collapse.

On the contrary, DINO requires the student model to learn global atmospheric context by utilizing small pixel patches of a very localized environment. Self-distillation through DINO is an ideal method of capturing the essence of haze, which has scale invariance and pervasiveness. Through its ability to capture local textures without having to identify structural features, DINO achieved success in identifying the 301-600 class with 59.38% accuracy.

In addition, the results on label efficiency confirm that SSL can be implemented effectively for addressing the AQI data bottleneck problem. DINO was able to achieve a classification rate of 30.17% with 1% of the total labeled training data, indicating that the model is capable of developing an inherent knowledge of the degradation in visual appearances during the pretraining stage. Indeed, the Silhouette scores that were found to be negative for all models accurately indicate the continuous aspect of air quality as opposed to discrete object classes that do not overlap.

8.2. Conclusion

In light of the crucial lack of data for training vision-based PM2.5 air quality estimation models, this paper explores the potentiality of recent SSL techniques as a means of avoiding extremely supervised methods. The experiments involved leveraging the PM25Vision dataset along with the EfficientNet-B0 model to analyze and compare bootstrapping (BYOL), redundancy reduction (Barlow Twins), and multi-crop self-distillation (DINO), assessing their ability to learn general atmospheric representation without having to rely on exhaustive labeled data.

The findings of this experiment unambiguously demonstrate the superiority of the self-distillation method embodied in DINO when processing continuous visual patterns such as atmospheric haze. Apart from the outstanding linear probing accuracy of DINO at 45.50%, it obtained a high fine-tuned accuracy of 68.36%, easily outperforming the highest-performing supervised algorithm. Most importantly, this study identified a key issue in the design of conventional non-contrastive SSL techniques: while both BYOL and Barlow Twins encountered

representation collapse in cases of extreme visual perturbation with zero percent performance in the dangerous range of 301-600 AQI, DINO managed to learn the formless haze and surpassed 59% accuracy in the same range.

In addition to that, DINO showed remarkable sample efficiency compared to full datasets of other SSL approaches through employing only 1% of labeled data available in the downstream task. Empirical support in the form of statistical testing for representation superiority ($p<0.05$) shows that the present research is able to prove that the prediction of global atmospheric context based on small patches of image pixels corresponds perfectly to physical laws governing light scattering in the atmosphere. Thus, the current research shows that self-distillation-based SSL constitutes a highly scalable, robust, and label-efficient framework for monitoring air pollution levels around the globe.

9. References

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